



- A watch which gains 10 s in 5 min was set correct at 9 am when the watch indicated 20 min past 7 O' clock, the same evening, the true time is **(Clock) NIMCET-2008**
(a) 7 pm (b) 7 : 40 pm (c) 7 : 10 pm (d) 8 pm
- A boy observes the reflection of a clock in a mirror. The time observed by the boy in the mirror is 3 h 45 min. what is the actual time shown in the clock? **(Clock) NIMCET-2008**
(a) 8 h 45 min (b) 9 h 45 min
(c) 8 h 15 min (d) 9 h 15 min
- A clock is set right at 8 a.m. The clock gains 10 minutes in 24 Hrs. What will be the right time when the clock indicates 1 p.m. on the following day ?
(Clock) NIMCET-2011
(a) 11.40 p.m. (b) 12.48 p.m.
(c) 12 noon (d) 10 p.m.
- Jimmy saw the time while going to the tennis court. He saw the hour hand is 20° away from 4. After he returned from tennis court, he noticed that the hour hand is 20° away from 4. If he took ten minutes to go to tennis court and he walked at the same speed while going to the tennis court and while returning, how much time did he spent at the tennis court?
(Clock) NIMCET 2013
(a) 60 min (b) 80 min
(c) 70 min (d) 50 min
- A watch gains 10 s in 5 min was set correct at 9:00 am. When the watch indicated 20 min past 7:00 pm in the same evening, the correct time is **(Clock) NIMCET-2014**
(a) 7:00 pm (b) 7:40 pm
(c) 7:10 pm (d) 8:00 pm
- How many times do the hour and the minute hands of a clock overlap in 24 h? **(Clock) NIMCET-2015**
(a) 24 (b) 22
(c) 26 (d) 20
- A clock is set right at 5 am the clock losses 16 minutes in 24 hours. What will be the correct time when the clock indicates 10 pm on the 3rd day? **(Clock) NIMCET-2016**
(a) 11 pm (b) 10: 45 pm
(c) 11: 15 pm (d) 12 pm
- At what time, in minutes, between 3 O' clock and 4 O' clock, both the needles will coincide each other? **(Clock) NIMCET-2018**
(a) $5\frac{1}{2}$ " (b) $12\frac{4}{11}$ " (c) $13\frac{4}{11}$ " (d) $16\frac{4}{11}$ "
- A clock is set right at 5 AM. The clock loses 16 min in 24 h. What will be the true time when the clock indicates 10 PM on 4th day? **(Clock) NIMCET-2019**
(a) 11:15 PM (b) 11:00 PM
(c) 12:00 PM (d) 12:30 PM
- It was 9.35 am in Garvita 's watch, which kept correct time, when Manya informed her that the last bus left the bus stop at 9.25 am. Manya' s watch is 5 min fast. The frequency of the bus is every 20 min. For how long Garvita must wait to catch the next bus? **(Clock) NIMCET-2020**
(a) 5 min (b) 10 min (c) 15 min (d) 20 min

- How many times in a day are the hands of a clock pointing opposite to each other? **(Clock) NIMCET-2021**
(a) 4 (b) 20 (c) 22 (d) 14
- At what time, in minutes, between 3 o' clock and 4o' clock both the needles will coincide each other? **(Clock) NIMCET-2022**
(a) $5\frac{1}{11}$ (b) $12\frac{4}{11}$ (c) $13\frac{4}{11}$ (d) $16\frac{4}{11}$
- At what time between 2 pm and 3 pm, will the hour and minute hands of a clock point in opposite directions (diametrically opposite)? **(Clock) NIMCET-2024**
(a) 2:45 pm (b) 2:44 pm
(c) $2:43\frac{9}{11}$ pm (d) $2:43\frac{7}{11}$ pm

ANSWER

1	a	2	c	3	b	4	a	5	a
6	b	7	b	8	d	9	b	10	a
11	c	12	d	13	d				

- (a)** As the watch gains 10 s in 5 min so it gains 2 min in 1 h. 20 min past 7 O'clock implies 10 h 20 min after 9 am, so actual time is 10 h ahead and 20 min is the gain. Hence, actual time is 7 pm.
- (c)** As by reflexion left hand becomes right hand and vice-versa, so in original clock hour hand is between 8 and 15 minute
- (b)** Time from 8 am of a particular day to 1 pm on the following day = 29 h. Now, the clock gains 10 min in 24 h, it means that 24 h 10 min of this clock is equal to 24 h of the correct clock. $145\frac{5}{6}$ h of this clock = 24 h of the correct clock

$$\frac{145\frac{5}{6}}{24} \times 24\text{h} = 145\frac{5}{6}\text{h}$$

$$= 28\text{h} 48\text{min}$$
It means that the clock in question is 12 min faster than the correct clock. Therefore, when clock indicates 1 pm the correct time will be 48 min past 12.
- (a)** Total movement of hour hand = $20^\circ + 20^\circ = 40^\circ$
So, hour hand covers $\frac{1^\circ}{2}$ in 1 min.
Also, hour hand covers 40° in 80 min.
Time duration to go tennis court and returning in (10+10) min
= 20 min.
 $\therefore$ Total spent time at the tennis court = (80 – 20) min
= 60 min
- (a)** Time gained in 5 min = 10 s
So, time gained in 60 min = $\frac{10}{5} \times 60$
= 120 s = 2 min
So, time gained in 1 h = 2 min
Time gained in 10 h = 2×10 min = 20 min
Now,
Present time = 7:20 pm
So, correct time is 7 pm.





6. (b) The hands of a clock coincide 11 times in every 12 h (Since, between 11 and 1, they coincide only once i.e. at 12 O'clock).

AM	PM
12:00	12:00
1:05	1:05
2:11	2:11
3:16	3:16
4:22	4:22
5:27	5:27
6:33	6:33
7:38	7:38
8:44	8:44
9:49	9:49
10:55	10:55

The hands overlap about 65 min, not every 60 min. So, the hand coincide 22 times in a day.

7. (b) Total hours from 5 am to 10 pm of 3rd day = 65h
The clock loses 16 min in 24 h.
i.e. 23 h 44 min of the clock = 24h of the correct clock
So, 65 h of clock = $\frac{24 \times 15 \times 65}{356}$ of the correct clock
= 65.73h
= 65h 44 min
So, correct time = 10:45pm (approx)
8. (d) At 3 O' clock both the hands of the clock are 15 min space apart.
To coincide, the minute hand must gain 15 min over the hour hand.
Now, 55 min are gained in 60 min.
 $\therefore$ 15 min would be gained in $\frac{60}{55} \times 15 = 16\frac{4}{11}$ min.
Hence, the needles will coincide after $16\frac{4}{11}$ min.
9. (b) Time from 5 am on a day to 10 pm on 4th day = 89h
Now 23 hours 44 min of this clock 24 hours of correct clock.
 $23\frac{44}{60} = 356/15$ h of this clock = 24 h of correct clock. 89 h
 $(24 \times \frac{15}{356} \times 89)$ h of correct clock.
= 90 h of correct clock.
so, the correct time is 11 pm.
10. (a) It was 9.35 AM is Garvita's watch, which kept correct time.
 $\Rightarrow$ Manya informed her that the last bus left the bus stop at 9.25 am. Manya's watch is 5 min fast.
 $\Rightarrow$ So, the actual time when the bus left the bus stop.
= 9.25 - 0 : 05 AM = 9:20 AM.
 $\Rightarrow$ The frequency of the bus is every 20 min.
So, the next bus will arrive at bus stop at 9:40 AM.
 $\Rightarrow$ As it was 9.35 AM in Garvita's watch, she should wait, 5 more minutes to catch the next bus.
 $\therefore$ The correct answer is options (a).
11. (c) 22 times in a day are the hands of a clock pointing opposite to each other.

12. (d) Here, $n = 3$ and $(n + 1) = 4$
According to the formula,
The two hands will coincide at $\frac{60n}{11}$ min past n or $\frac{60 \times 3}{11}$ min past 3 or $16\frac{4}{11}$ min past 3.

13. (d)
Min + $\frac{\text{min}}{11}$
 $40 + \frac{40}{11}$
 $43\frac{7}{11}$